CSE400 M2 GROUP- 8 Climate

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Project Context and Motivation

Rain forecasting is uncertain in nature because :-

- The atmospheric systems are chaotic and nonlinear.
- Radar measurements are characterized by noise in observations.
- Minor inconveniences grow with time.

Although we may not know that the future rain will fall exactly in this or that distribution, we can still reason about:

Its acting tendencies, its changeability and limits on deep extremes.

This can be put in terms of the concentration inequalities argument where partial information (mean and variance) can be used to bound rare events.

Therefore, rainfall forecasting in our project is viewed as a concept of uncertainty, rather than deterministic prediction.

Project Objective and Scope

Objective:

To model rainfall as a random process and approximate:

- Distribution of rainfall in future,
- Probability of extreme occurrences,
- The spreading of uncertainty over time.

Modeling Goals:

- Determine important random variables of the rainfall evolution system.
- Define right probability distributions.
- Mimic occurrences by simulating the stochastic system.
- Produce PDF and CDF plot in order to develop intuition.

Our daily expectation is not actually to predict one value.

We strive to describe the whole distribution of the possible outcomes.

This transforms the issue into a given probabilistic inference.

Project System Overview

The system can be perceived as stochastic state-space model:

- Rainfall is the state variable which is transformed logarithmically.
- The state passed through an autoregressive mechanism changes with time.
- Certain innovations bring about noise.
- Spatial uncertainty is brought about by motion perturbations.
- Ensemble simulations are an approximation of the predictive distribution.

This is a layered stochastic modeling in which uncertainty is injected, developed and measured out probabilistically.

Sources of Uncertainty

There are multiple points of uncertainty:

1.Measurement Noise

Radar data is noisy. Recorded precipitation is already questionable.

2.Environmental Stochasticity

Cells of rain develop and dissipate randomly.

3.Motion Field Uncertainty

Advectional models are based on the assumption that there is constant motion which is not real.

4.Model Approximation

Assumption of logarithmic transformation, this occurs when the underlying distribution follows a Gaussian process rather than a normal distribution; the former is skewed outward.

Parameter estimation error

Zero consideration of spatial dependence.

In probabilistic terms:

At every step we have more variance that was generated, and propagation increases uncertainty.

Key Random Variables

1.Rainfall Intensity (Raw Rainfall)

- This is a physical rate of precipitation.
- It is modeled as log-normal.

That means:

- It is positively skewed.
- Extreme values are possible.
- The tail heavy rainfall occurrences are at the tail.
- The skewness indicates actual rainfall action.

2.Log-Transformed Rainfall

This is the fundamental modeling variable.

After transformation:

- The distribution turns to be nearly Gaussian.
- Mean is a measure of central tendency of transformed space rainfall.
- Variance shows the level of uncertainty.

Because it is Gaussian:

- Mean and variance completely characterize it.
- Calculation of analytical probability is possible.
- Noise of innovation can be put in place steadily.
- This forms the basis of the stochastic structure.

3.Innovation Noise Term

This is a model that has random succession of time steps.

Conceptually:

Though the amount of rainfall yesterday is known, the amount of rainfall in the morning tomorrow cannot be known accurately.

This innovation term: Has zero mean, adds variance, and motives uncertainty increase in the long run.

It makes sure that the model does not degenerate to its deterministic behaviour.

4.Motor Perturbation Variables

Estimated velocity fields are used to advect rainfall fields.

Yet there is something wrong with velocity estimates. Parallel and perpendicular perturbation are presented and modelled by a laplace distribution.

Why Laplace?

Since it is more tailed than Gaussian.

This means:

The probability of large motion errors is greater than that of assuming a normal distribution.

This represents the actual displacement uncertainty.

5.Exceedance Indicator Variable

This is a binary variable.

It answers:

Natural climate Change: Does rainfall increase above a limit?

This turns incessant uncertainty into a variable of decision making.

It allows:

- Computation of the probability of flood risk,
- Decision support (operations).
- This is the place of execution of probabilistic modeling.

Probabilistic Models and Assumptions

Core Modeling Assumptions:

- Following log transformation rainfall is almost Gaussian.
- Temporal evolution assumes the form of an autoregressive process.
- Innovation noise, mean, and variance are both not equal to zero.
- The perturbations in motion take a heavy-tailed Laplace distribution.
-

Interpretation of the Model:

- The average is projected rainfall behaviour.
- The variance is a degree of uncertainty.
- Noise added by innovation brings about randomness on a time step basis.
- Motions perturbations explain errors of displacement.
- The uncertainty increases with the further length of forecast.

MOTION PERTURBATION (LAPLACE)

Random Variable:

$\Delta X \sim \text{Laplace}(0, b)$

Probability Density Function:

$f(x) = (1 / 2b) \exp(-|x| / b)$

Cumulative Distribution Function:

For $x < 0$:

$F(x) = (1/2) \exp(x/b)$

For $x \geq 0$:

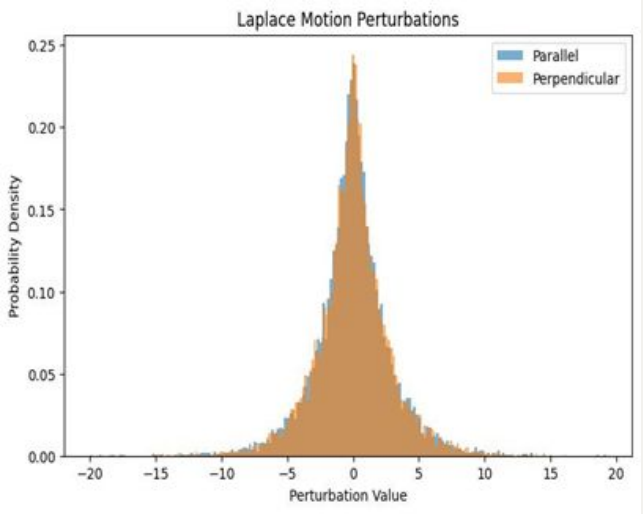
$F(x) = 1 - (1/2) \exp(-x/b)$

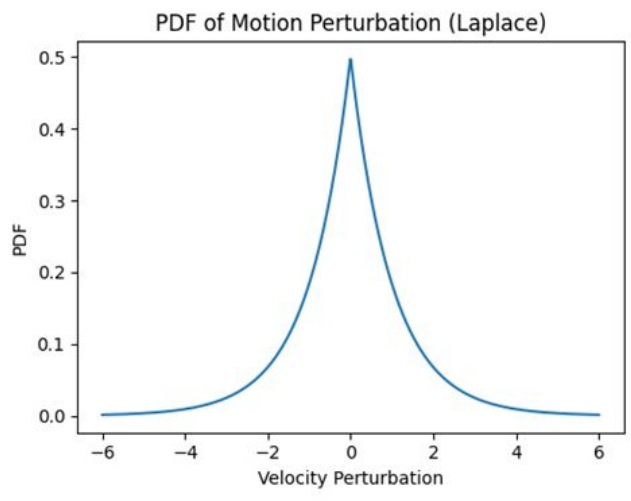
Mean:

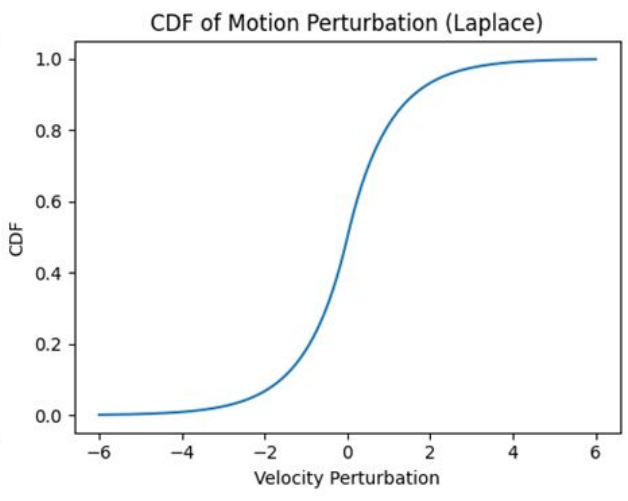
$E[\Delta X] = 0$

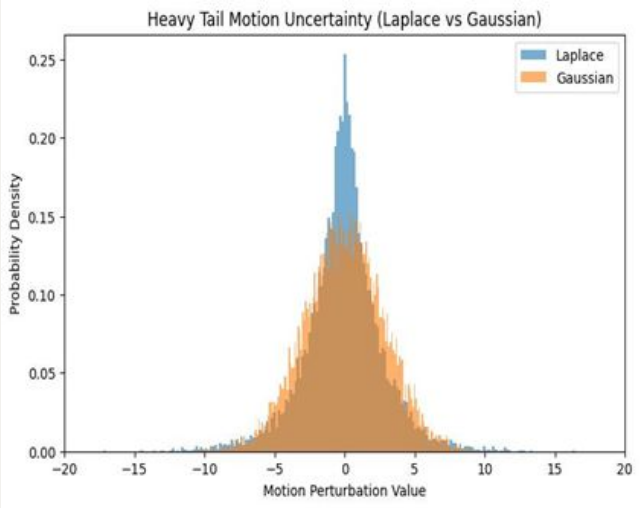
Variance:

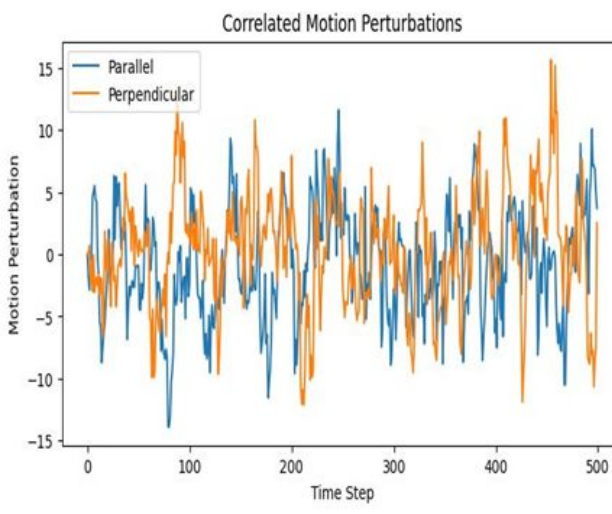
$\text{Var}(\Delta X) = 2b^2$











RAINFALL INTENSITY (LOG-NORMAL)

Random Variable:

$R \sim \text{Log-Normal}(\mu, \sigma^2)$

Log Transformation:

$Y = \log(R)$

$Y \sim N(\mu, \sigma^2)$

Probability Density Function:

$f_R(r) = (1 / (r \sqrt{(2\pi\sigma^2)})) \times \exp(-(\log r - \mu)^2 / (2\sigma^2)), \quad r > 0$

Cumulative Distribution Function:

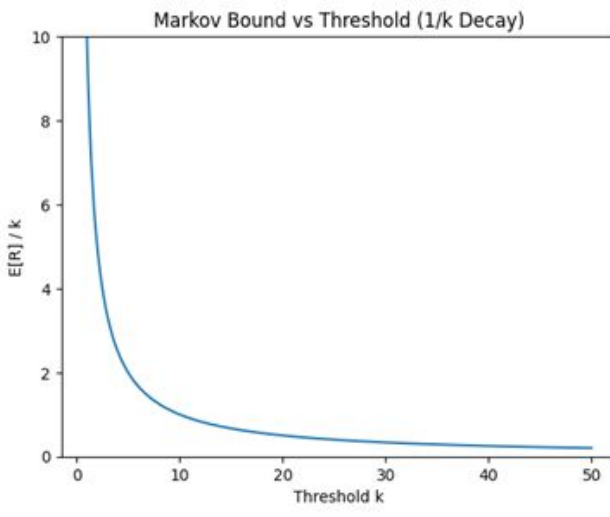
$F_R(r) = P(R \leq r) = \Phi((\log r - \mu) / \sigma)$

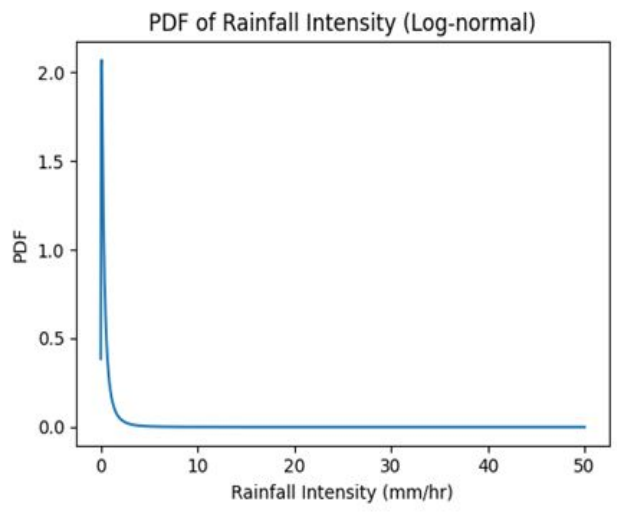
Mean:

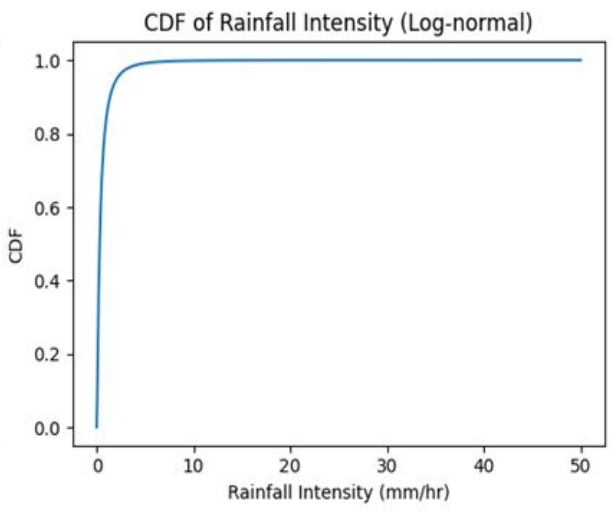
$E[R] = \exp(\mu + \sigma^2 / 2)$

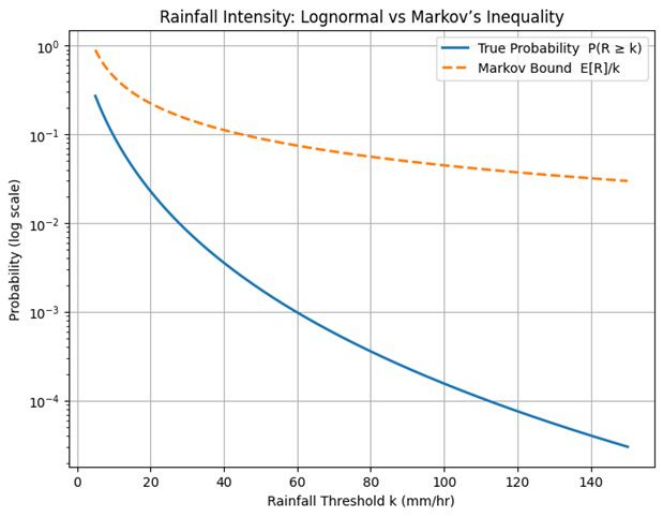
Variance:

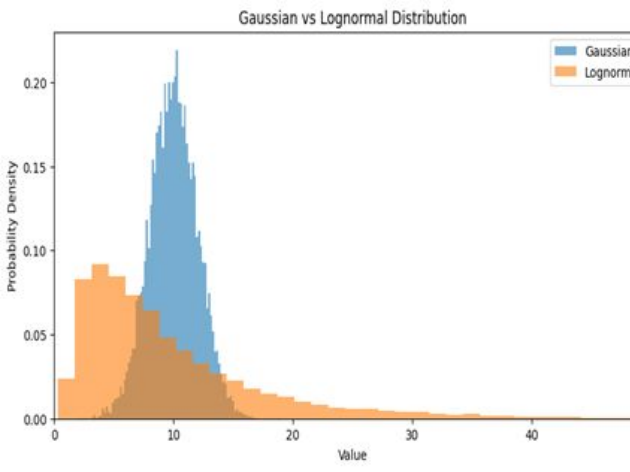
$Var(R) = (\exp(\sigma^2) - 1) \exp(2\mu + \sigma^2)$











LOG-TRANSFORMED RAINFALL (GAUSSIAN)

Random Variable:

$Y = \log(R)$

Distribution:

$Y \sim N(\mu, \sigma^2)$

Probability Density Function:

$f_Y(y) = (1 / \sqrt{(2\pi\sigma^2)}) \times \exp(-(y - \mu)^2 / (2\sigma^2))$

Cumulative Distribution Function:

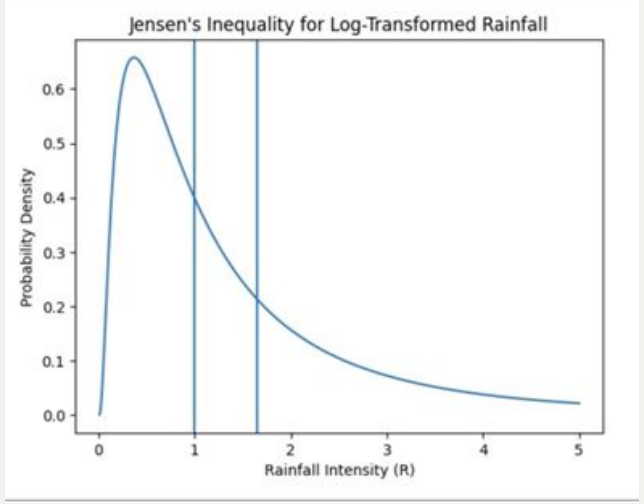
$F_Y(y) = P(Y \leq y) = \Phi((y - \mu) / \sigma)$

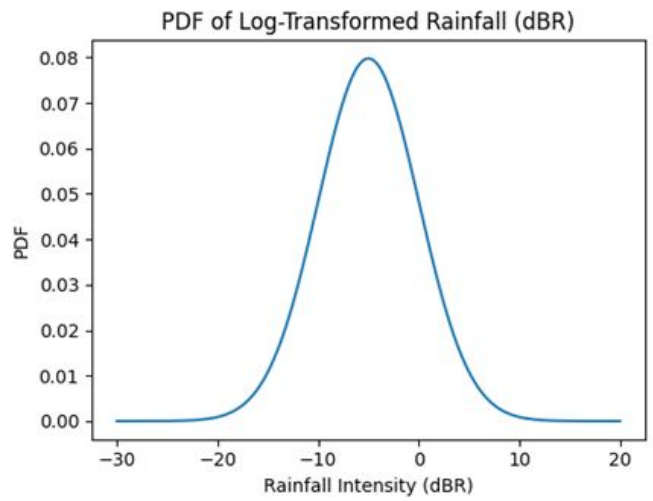
Mean:

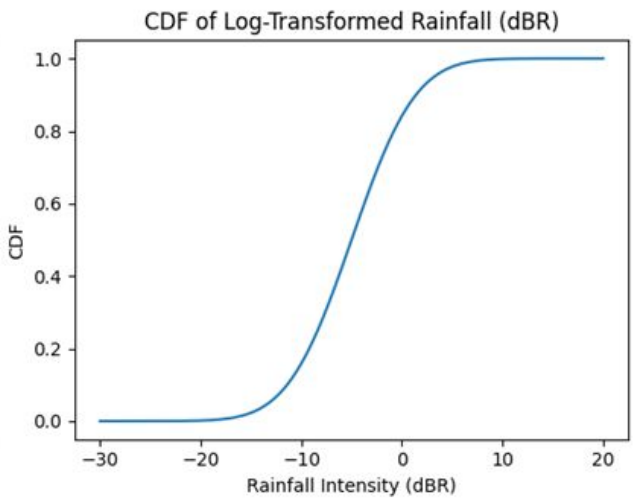
$E[Y] = \mu$

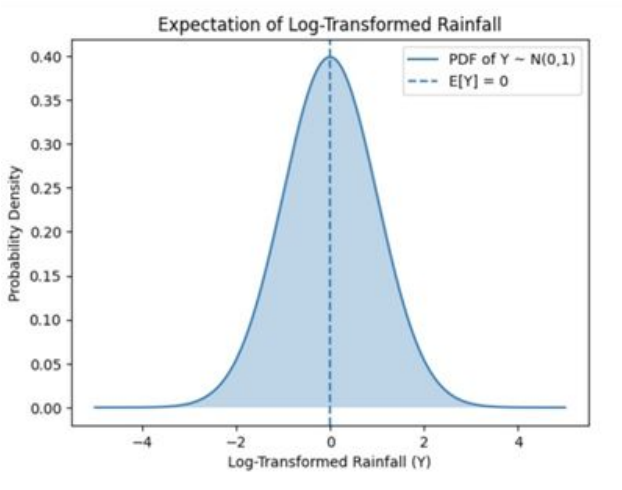
Variance:

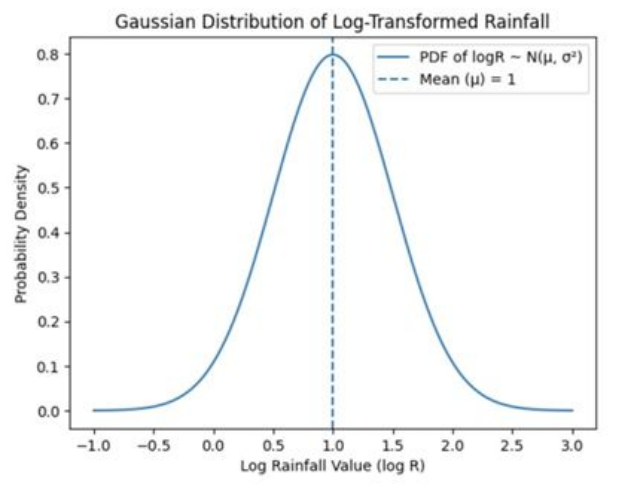
$Var(Y) = \sigma^2$











STOCHASTIC NOISE (GAUSSIAN)

Random Variable:

$\varepsilon \sim N(0, \sigma^2)$

Probability Density Function:

$f(\varepsilon) = (1 / \sqrt{(2\pi\sigma^2)}) \times \exp(-\varepsilon^2 / (2\sigma^2))$

Cumulative Distribution Function:

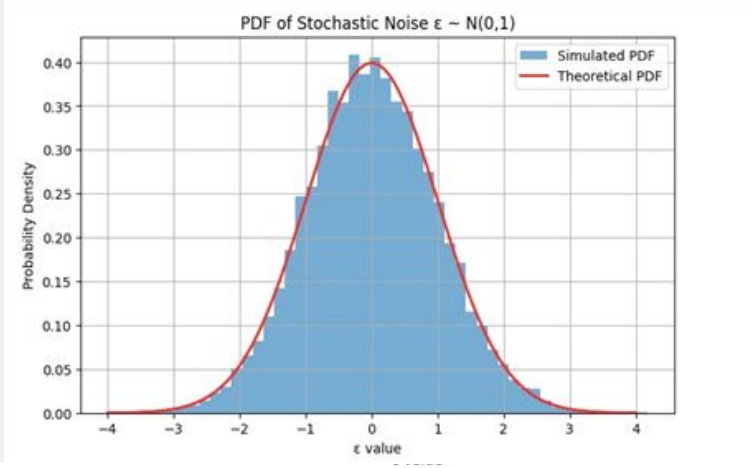
$F(x) = P(\varepsilon \leq x) = \Phi(x / \sigma)$

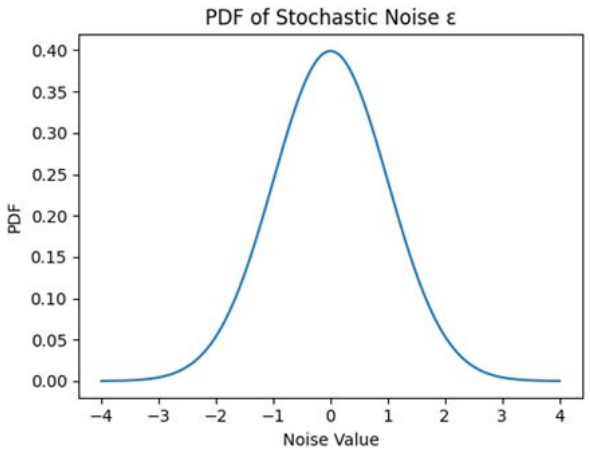
Mean:

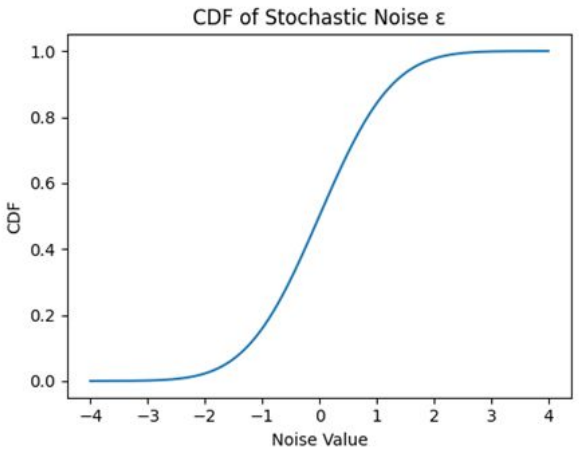
$E[\varepsilon] = 0$

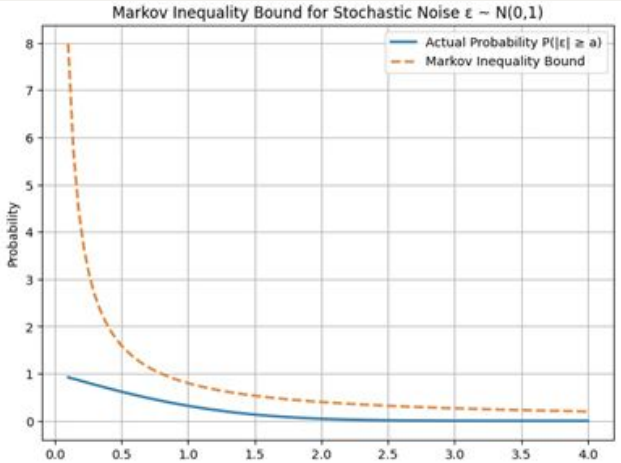
Variance:

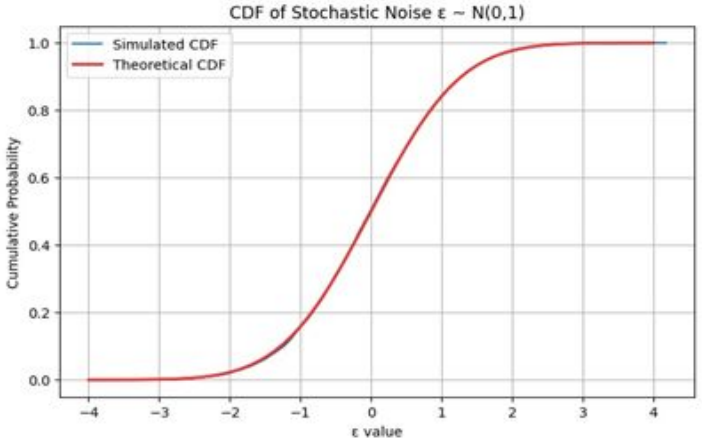
$Var(\varepsilon) = \sigma^2$











MONTHLY AVERAGE RAINFALL (CHEBYSHEV BOUND)

Random Variable:

$\varepsilon \sim N(0, \sigma^2)$

Probability Density Function:

$f(\varepsilon) = (1 / \sqrt{(2\pi\sigma^2)}) \times \exp(-\varepsilon^2 / (2\sigma^2))$

Cumulative Distribution Function:

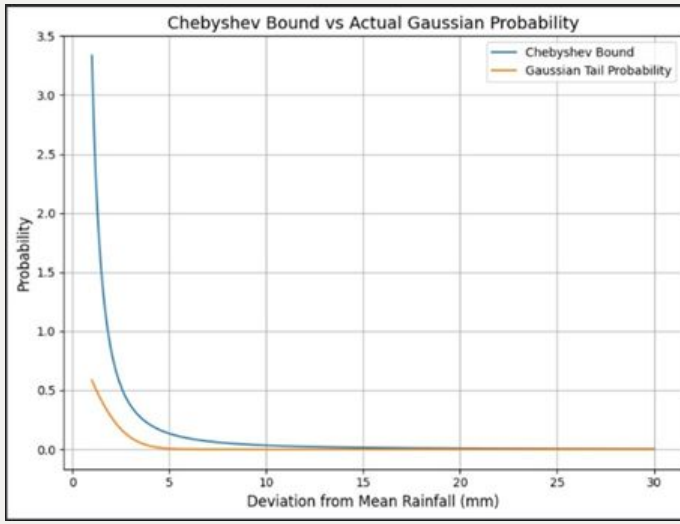
$F(x) = P(\varepsilon \leq x) = \Phi(x / \sigma)$

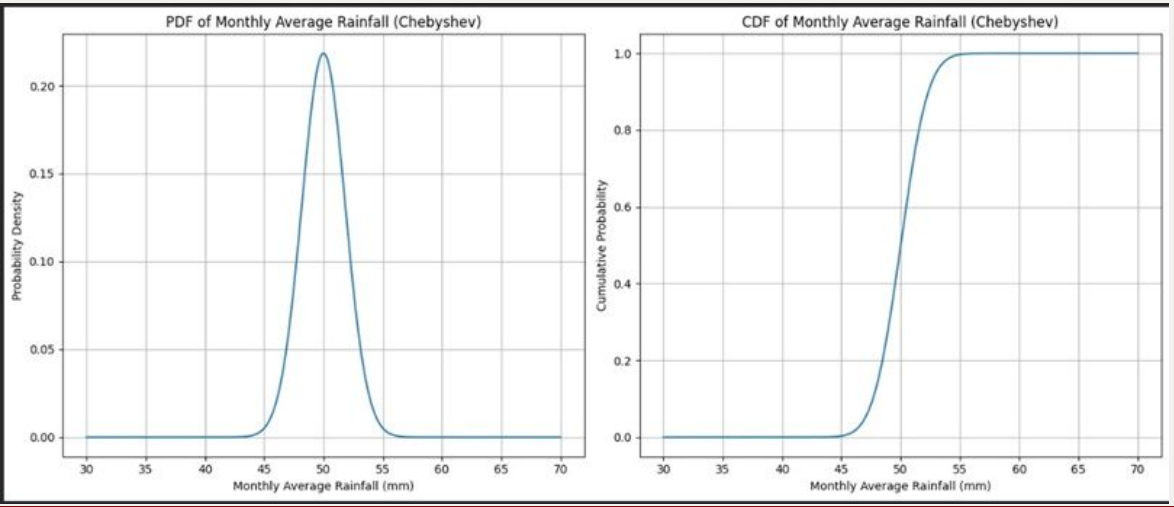
Mean:

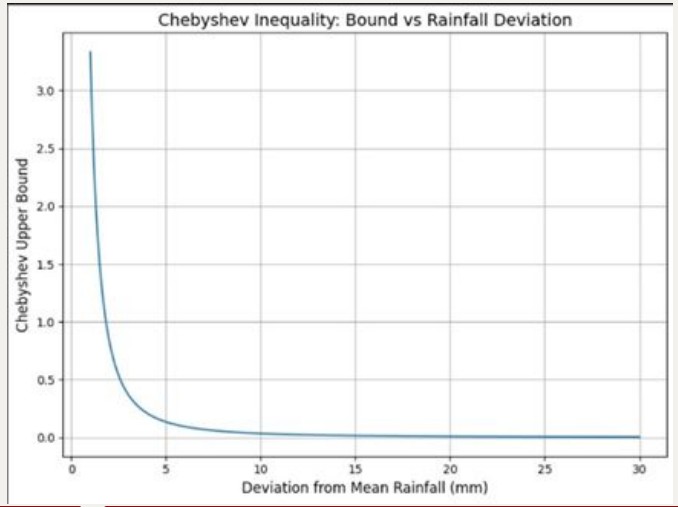
$E[\varepsilon] = 0$

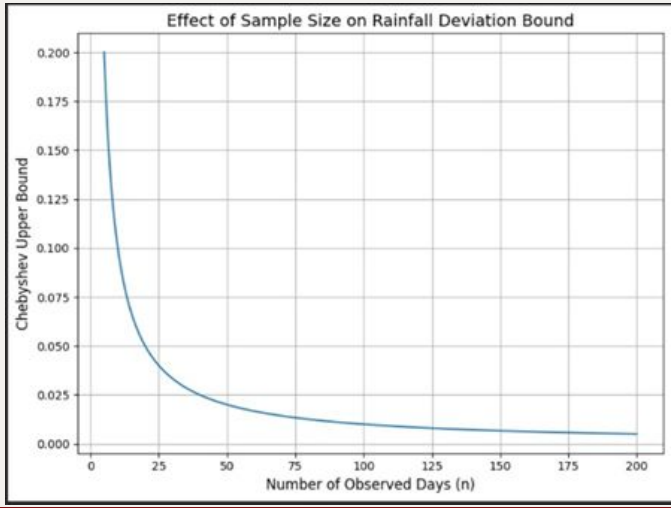
Variance:

$Var(\varepsilon) = \sigma^2$









Probabilistic Reasoning and Inference Logic

Rainfall forecasting is done as a stochastic process.

Key reasoning:
- Rain is unpredictable and discontinuous.
- Measurements contain noise.
- Small variations increase with time.

Rather than making a prediction of a single value, that is, we define the entire probability distribution.

The extreme rainfall is limited with the following:
- Markov Inequality
- Chebyshev Inequality

Operationalizing Probabilistic Reasoning

Rainfall Intensity:
$R \sim \text{Log-Normal}(\mu, \sigma^2)$

Log Transformation:
$Y = \log(R)$
$Y \sim N(\mu, \sigma^2)$

Motion Perturbation:
$\Delta X \sim \text{Laplace}(0, b)$

Stochastic Noise:
$\varepsilon \sim N(0, \sigma^2)$

Exceedance Indicator:
$I = 1 \text{ if } R \geq k$
$I \sim \text{Bernoulli}(p)$

Current Limitations and Conceptual Gaps

1. Measurement Noise is one of the issues in our project. It is usually Gaussian.
2. We also have to deal with Environmental Randomness.
3. Sometimes Motion Field Errors. They follow a Laplace pattern.
4. Then there is Model Approximation, which isn't perfect.

Over time we are less sure about our measurements due to Uncertainty, in Measurement Noise and Model Approximation.

Summary of Group-Level Understanding

- We think of rainfall as something that might happen not something that will happen for sure.
- When we talk about rainfall we do not just say it is an amount we say it could be one of many amounts.
- The log-normal way of thinking about rainfall is good because it shows us that most of the time rainfall is not very much. Sometimes it can be a lot.
- The Laplace way of thinking about rainfall is good for showing us that sometimes rainfall can be very very heavy.
- We use something called Gaussian noise to think about how much we do not know about what will happen with the rainfall.
- We use math to figure out the worst case scenario, for rainfall so we can be prepared for really bad storms.